

Chapter 10. Singularly Perturbed Impulsive-Switched Systems w/ Time Delay.

10.1 Problem Formulation.

$$(10.1) \begin{cases} \dot{x} = f_{\sigma(t)}(t, x, x_t, z, z_t), & t \neq t_k \\ \varepsilon \dot{z} = g_{\sigma(t)}(t, x, x_t, z, z_t), & t \neq t_k \\ \Delta x = B_k x(t_k^-), & t = t_k \\ \Delta z = C_k z(t_k^-), & t = t_k \end{cases} \quad \begin{array}{l} \text{Impulse.} \\ \downarrow \\ \text{Since } x(t_k^-) = \lim_{t \rightarrow t_k^-} x(t) \end{array}$$

where $\sigma: [t_0, \infty) \rightarrow J = \{1, \dots, N\} = J_u \cup J_s$

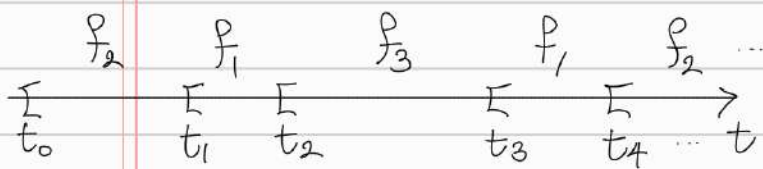
piecewise const. fn.

of subsystems

$\sigma(t) := i_k$ if $t \in [t_{k-1}, t_k)$: discontinuous at each t_k .

$\Rightarrow \{t_k\}_{k=1}^{\infty}$: seq. of impulsive-switching times.

s.t. $t_{k-1} < t_k$, $\lim_{k \rightarrow \infty} t_k = \infty$



$i_1 = 2, i_2 = 1, i_3 = 3$
 $i_4 = 1, \dots$
 On 4th subinterval $[t_3, t_4)$,
 1st subsystem
 is activated.

* $T^+(t_0, t)$ ($T^-(t_0, t)$): total activation time of unstable (stable) subsystem over the time interval $[t_0, t)$

Note $T^+(t_0, t) + T^-(t_0, t) = \text{the length of } [t_0, t) = t - t_0$.

* Assumption

① $x(t_k^+) = x(t_k^-)$, $z(t_k^+) = z(t_k^-)$: the solution of the system is right continuous.

② f_{i_k}, g_{i_k} are smooth.

Thm 10.1

trivial solution of (10.2) is exponentially stable if

(A1) $i_k \in J_u \Rightarrow A_{ii_k}$ has eigenvalues w/ positive real parts

$i_k \in J_s \Rightarrow A_{ii_k}$ has " w/ negative "

(A2) $\forall K \in \mathbb{N}, t \in [t_{K-1}, t_K), \exists a_{\square}, b_{\square} > 0$ s.t. (chap. 9)

① $2x^T P_{i_k} [A_{12i_k} + \dots] \leq a_{i_k} \|x\|^2 + a_{2i_k} \|x_t\|_t^2 + \dots$ " I_m " I_n

② P_{i_k}, P_{2i_k} : Positive-definite matrix. For given Q_{i_k}, Q_{2i_k}

$$\lambda_{1m} = \min \{ \lambda_{\min}(P_{i_k}) \mid k \in \mathbb{N} \}$$

$$\lambda_{1M} = \max \{ \lambda_{\max}(P_{i_k}) \mid k \in \mathbb{N} \}$$

$$\Rightarrow \mu = \max \left\{ \frac{\lambda_{1M}}{\lambda_{1m}}, \frac{\lambda_{2M}}{\lambda_{2m}} \right\}$$

(A3) : conditions are designed to apply Lem 2.3 and Lem 2.4,

$\exists r^* > 0$ s.t. $A_{ii_k} - r^* I$: Hurwitz ($\forall k$)

(A4) : $\lambda^+ = \max \{ \xi_{i_k}^{ksi} \mid i_k \in J_u \}$, $\lambda^- = \min \{ \xi_{i_k}^{zetaa} \mid i_k \in J_s \}$

growth rate

decay rate of subsystem

defined in Lem 2.3 and 2.4 (P.14)

$$(10.3) : \exists \lambda^* \in (0, \lambda^-) \text{ s.t. } \inf_{t \geq t_0} \frac{T^-(t_0, t)}{T^+(t_0, t)} \geq \frac{\lambda^+ + \lambda^*}{\lambda^- - \lambda^*}$$

(10.4) (i) for $k \in \mathbb{N}$ satisfying $i_k \in J_u$,

$$\ln \mu(\alpha_k + \beta_k + r_k + \gamma_k) - v(t_k - t_{k-1}) \leq 0$$

$$\Rightarrow \mu(\alpha_k + \beta_k + r_k + \gamma_k) \leq e^{v(t_k - t_{k-1})}$$

(ii) for $k \in \mathbb{N}$ satisfying $i_k \in J_s$,

"

Proof

$$\begin{aligned} V_i(t) &:= V_i(x(t)) = x^T(t) P_{1i} x(t) \\ W_i(t) &:= W_i(z-h_i(t)) = (z-h_i(t))^T P_{2i} (z-h_i(t)) \\ \text{where } h_i(t) &:= -B_{2i}^{-1} [A_{21} x + A_{22} z_t] \\ \text{so that } B_{2i} h_i &= A_{21} x + A_{22} z_t. \end{aligned}$$

defined in A2.

(*) (i) If $i_k \in J_u, t \in (t_{k-1}, t_k)$ $\forall t_{k-1} \in (t_{k-1}, t_k)$, $\sup_{t_{k-1}-\tau \leq s \leq t_k} \|V_{i_k}(s)\|$

this ineq. holds for $\forall t \in [t_{k-1}, t_k)$

$$\begin{pmatrix} \dot{V}(t) \\ \dot{W}(t) \end{pmatrix} \leq \underbrace{\begin{pmatrix} \square & \Delta \\ 0 & \star \end{pmatrix}}_{\tilde{A}_i} \begin{pmatrix} V(t) \\ W(t) \end{pmatrix} + \underbrace{\begin{pmatrix} \square' & \Delta' \\ 0' & \star' \end{pmatrix}}_{\tilde{B}_i} \begin{pmatrix} \|V_{i_k}\|_{\tau} \\ \|W_{i_k}\|_{\tau} \end{pmatrix}$$

A3(i), Lemma 2.4

$\Rightarrow \exists \xi_i > 0$ s.t.

$$V_{i_k}(t) \leq (\|V_{i_k, t_{k-1}}\|_{\tau} + \|W_{i_k, t_{k-1}}\|_{\tau}) e^{\xi_i(t-t_{k-1})}$$

$\sup_{t_{k-1}-\tau \leq s \leq t_{k-1}} \|V_{i_k}(s)\|$

right
the continuity of $x(t), z(t)$
ensures the right continuity
of V and W .

$$W_{i_k}(t) \leq (\|V_{i_k, t_{k-1}}\|_{\tau} + \|W_{i_k, t_{k-1}}\|_{\tau}) e^{\xi_i(t-t_{k-1})}$$

for $\forall t \in [t_{k-1}, t_k)$

(**) (ii) If $i_k \in J_s$, A3(ii), Lemma 2.3 $\exists \xi_i > 0$ s.t.

zeta $\dot{x}(t) = \begin{matrix} \text{hatched box} \end{matrix}$ $t \neq t_k$.

$$V_{i_k}(t) \leq (\|V_{i_k, t_{k-1}}\|_{\tau} + \|W_{i_k, t_{k-1}}\|_{\tau}) e^{-\xi_{i_k}(t-t_{k-1})}$$

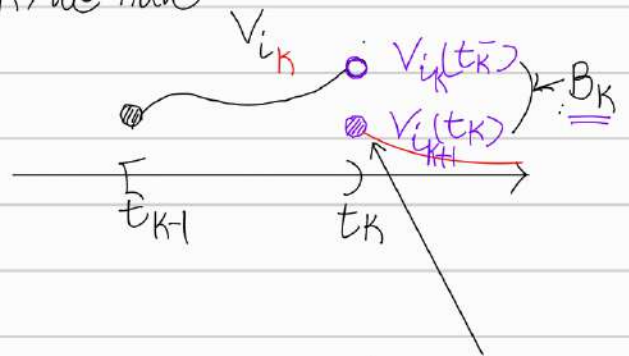
$$W_{i_k}(t) \leq (\|V_{i_k, t_{k-1}}\|_{\tau} + \|W_{i_k, t_{k-1}}\|_{\tau}) e^{-\xi_{i_k}(t-t_{k-1})}$$

for $\forall t \in [t_{k-1}, t_k)$

At the impulsive-switching moment $t = t_k$, we have

$$V_{i_{k+1}}(t_k) = \alpha(t_k)^T P_{i_{k+1}} \alpha(t_k)$$

$$= [(I + B_k) \alpha(t_{k-})]^T P_{i_{k+1}} [(I + B_k) \alpha(t_{k-})]$$



$$\Delta \alpha = \alpha(t_k) - \alpha(t_{k-}) = B_k \cdot \alpha(t_{k-})$$

$$\Leftrightarrow \alpha(t_k) = \alpha(t_{k-}) + B_k \cdot \alpha(t_{k-}) = (I + B_k) \alpha(t_{k-})$$

right conti.
of α, z .

$$= \alpha(t_{k-})^T [(I + B_k)^T P_{i_{k+1}} (I + B_k)] \alpha(t_{k-})$$

$$\leq \lambda_{\max} [(I + B_k)^T P_{i_{k+1}} (I + B_k)] \cdot \|\alpha(t_{k-})\|^2$$

$$\leq \lambda_{\max} [(I + B_k)^T (I + B_k)] \lambda_{\max}(P_{i_{k+1}}) \|\alpha(t_{k-})\|^2$$

$$\leq \mu \lambda_{\max}^2(I + B_k) \cdot \lambda_{\min} \|\alpha(t_{k-})\|^2 \leq \mu \lambda_{\max} \leq \mu \cdot \lambda_{\min}$$

$$\mu = \max \left\{ \frac{\lambda_{1M}}{\lambda_{1m}}, \frac{\lambda_{2M}}{\lambda_{2m}} \right\} \geq \frac{\lambda_{1M}}{\lambda_{1m}}$$

$$\leq \alpha_k \cdot \lambda_{\min}(P_{i_k}) \cdot \|\alpha(t_{k-})\|^2 \leq \alpha_k \cdot \alpha(t_{k-})^T P_{i_k} \alpha(t_{k-})$$

this term is completely $V_{i_k}(t_k)$
determined by $B_k \Rightarrow V_{i_k}(t_{k-}) \xleftrightarrow{B_k} V_{i_{k+1}}(t_k)$

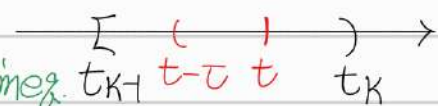
Similarly, we can show

$$W_{i_{k+1}}(t_k) \leq \gamma_k W_{i_k}(t_{k-}) + \beta_k V_{i_k}(t_{k-}) + \psi_k \|V_{i_k}(t_{k-})\|_{\mathcal{U}}$$

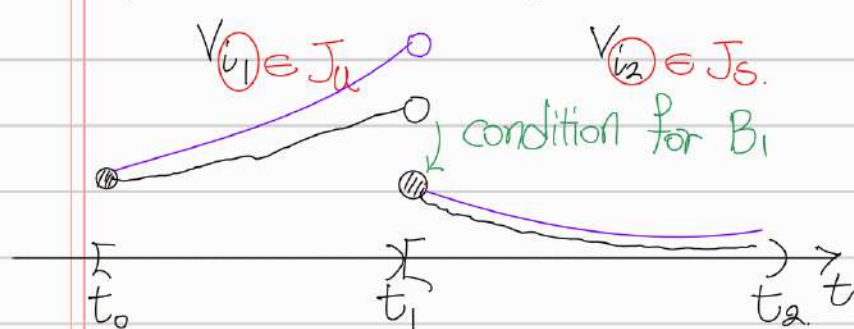
we want to control this using the information before the time t_k .

$$\lim_{t \rightarrow t_k} \|V_{i_k}(t)\|_{\mathcal{U}} \leq \sup_{t_{k-} \leq s \leq t_k} \|V_{i_k}(s)\|$$

defs of $\gamma_k, \beta_k, \psi_k$ and Cauchy-Schwartz ineq. are used.



Let's consider the case where we run an unstable subsystem on the first interval and a stable one on the second interval.



$$\sigma: [t_0, \infty) \rightarrow \mathcal{J}$$

$$\sigma([t_{k-1}, t_k]) = i_k$$

$$\{1, \dots, N\}$$

$$\|(V_{i_2})_{t_1}\|_{\mathcal{T}} := \sup_{\substack{t_0 \leq t_1 - \tau \leq t \leq t_1}} \|V_{i_2}(t)\|$$

$$t_1 - t_0 \geq \inf(t_k - t_{k-1}) \geq \tau > 0$$

$$\text{From def of } \mu := \max\left\{\frac{\lambda_{1M}}{\lambda_{1m}}, \frac{\lambda_{2M}}{\lambda_{2m}}\right\}$$

$$V_{i,j} \in \mathcal{J} = [N],$$

$$V_i(t) \leq \mu \cdot V_j(t)$$

$$\text{for } \forall t \in [t_0, \infty)$$

$$\textcircled{1} \text{ If } t \in [t_0, t_1) \Rightarrow \|V_{i_2}(t)\| = V_{i_2}(t) \leq \mu \cdot V_{i_1}(t)$$

not matched

$$\leq \mu \cdot (\|(V_{i_1})_{t_0}\|_{\mathcal{T}} + \|(W_{i_1})_{t_0}\|_{\mathcal{T}}) \cdot e^{\xi_1(t-t_0)}$$

by $\otimes$

$$\leq \mu \cdot (\|(V_{i_1})_{t_0}\|_{\mathcal{T}} + \|(W_{i_1})_{t_0}\|_{\mathcal{T}}) \cdot e^{\xi_1(t_1-t_0)}$$

$\odot$ $t_0 \leq t < t_1$

$$\textcircled{2} \text{ If } t = t_1, \|V_{i_2}(t_1)\| = V_{i_2}(t_1) \leq \alpha_K V_{i_1}(t_1)$$

matched well

$$\leq \alpha_K (\|(V_{i_1})_{t_0}\|_{\mathcal{T}} + \|(W_{i_1})_{t_0}\|_{\mathcal{T}}) e^{\xi_1(t_1-t_0)}$$

≥ 1 $\odot$

$$\Rightarrow \|(V_{i_2})_{t_1}\|_{\mathcal{T}} := \sup_{t_1 - \tau \leq t \leq t_1} \|V_{i_2}(t)\| \leq \alpha_1 \mu (\|(V_{i_1})_{t_0}\|_{\mathcal{T}} + \|(W_{i_1})_{t_0}\|_{\mathcal{T}}) e^{\xi_1(t_1-t_0)}$$

Similarly, we can show $\|(W_{i_2})_{t_1}\| \leq \mu (\beta_1 + \gamma_1 + \psi_1) (\|(V_{i_1})_{t_0}\|_{\mathcal{T}} + \|(W_{i_1})_{t_0}\|_{\mathcal{T}}) e^{\xi_1(t_1-t_0)}$

From this point, I have made some modifications

Generally, we have $\leq \prod_{i_k \in J_u} e^{\gamma(t_k - t_{k-1})}$

$\forall t > t_0$,
 $\exists! n \in \mathbb{N}$ s.t.
 $t \in [t_{n-1}, t_n)$

$$V_{i_n}(t) \leq \prod_{\substack{1 \leq k < n \\ i_k \in J_u}} M \cdot (\alpha_k + \beta_k + \gamma_k + \psi_k) e^{\xi_k(t_k - t_{k-1})} \leq \lambda^+ \prod_{i_l \in J_s} e^{\gamma(t_l - t_{l-1})} \leq -\lambda^-$$

$$\times \prod_{\substack{1 \leq l < n \\ i_l \in J_s}} M \cdot (\alpha_l + \beta_l + \gamma_l + \psi_l) e^{\xi_l(t_l - t_{l-1})} \times \prod_{i_l \in J_s} e^{\gamma(t_l - t_{l-1})}$$

$\xi_n(t - t_{n-1})$ if $i_n \in J_u$
 $-\xi_n(t - t_{n-1})$ if $i_n \in J_s$

$$\times (\|V_{i_1}\|_{t_0} + \|W_{i_1}\|_{t_0}) e^{\xi_n(t - t_{n-1})}$$

$\textcircled{A4} \leq e^{\gamma(t_{n-1} - t_0)} \times e^{\lambda^+ T^+(t_0, t)} \times e^{-\lambda^- T^-(t_0, t)} \times \boxed{\text{shaded box}}$

$\leq e^{\gamma(t - t_0)} \times e^{-\lambda^*(t - t_0)} \times \boxed{\text{shaded box}}$

$\textcircled{A4} \lambda^+ T^+(t_0, t) - \lambda^- T^-(t_0, t) \leq -\lambda^*(t - t_0)$

$\textcircled{7}: \forall t > t_0, \frac{T^-(t_0, t)}{T^+(t_0, t)} \geq \inf_{s \geq t_0} \frac{T^-(t_0, s)}{T^+(t_0, s)} \stackrel{\textcircled{A4}}{\geq} \frac{\lambda^+ + \lambda^*}{\lambda^- - \lambda^*}$

$\Rightarrow$ Since $\lambda^- - \lambda^* > 0$ ($\lambda^* \in (0, \lambda^-)$),

$$\lambda^- T^- - \lambda^* T^+ \geq T^+ \lambda^+ + T^- \lambda^*$$

$$\Leftrightarrow \lambda^+ T^+(t_0, t) - \lambda^- T^-(t_0, t) \leq -\lambda^* (T^-(t_0, t) + T^+(t_0, t))$$

$\leq e^{-(\lambda^* - \gamma)(t - t_0)} \times (\|V_{i_1}\|_{t_0} + \|W_{i_1}\|_{t_0}) e^{t - t_0}$

$\parallel$ by def.

$$\Rightarrow \|q(t)\|^2 \leq \frac{1}{\lambda_{\min}} V_{i_n}(t) \leq \frac{1}{\lambda_{\min}} \times \boxed{\text{shaded box}}$$

$\Rightarrow \exists K_1 > 0$ s.t. $\|q(t)\| \leq K_1 (\|q_{t_0}\|_{\mathcal{U}} + \|z_{t_0}\|_{\mathcal{U}}) e^{-(\lambda^* - \gamma)(t - t_0)/2}$

Similarly, we have

$$V_i(s) = x(s)^T P_i x(s).$$

$$W_{in}(t) \leq (\|(V_i)_{t_0}\|_{\mathcal{L}} + \|(W_i)_{t_0}\|_{\mathcal{L}}) e^{-(\lambda^* - \nu)(t-t_0)}$$

n th time interval $[t_{n-1}, t_n) \ni t$.

By using the fact that

$$\|z\| - \|h_{in}\| \leq \|z - h_{in}\| \leq \frac{1}{\sqrt{\lambda_{2m}}} W_{in}^{1/2}$$

We can see $\exists K_2$ s.t.

$$\|z(t)\| \leq K_2 (\|x_{t_0}\|_{\mathcal{L}} + \|z_{t_0}\|_{\mathcal{L}}) e^{-(\lambda^* - \nu)(t-t_0)/2}.$$

$$\therefore \|x(t)\| + \|z(t)\| \leq K (\|x_{t_0}\|_{\mathcal{L}} + \|z_{t_0}\|_{\mathcal{L}}) e^{-(\lambda^* - \nu)(t-t_0)/2}.$$